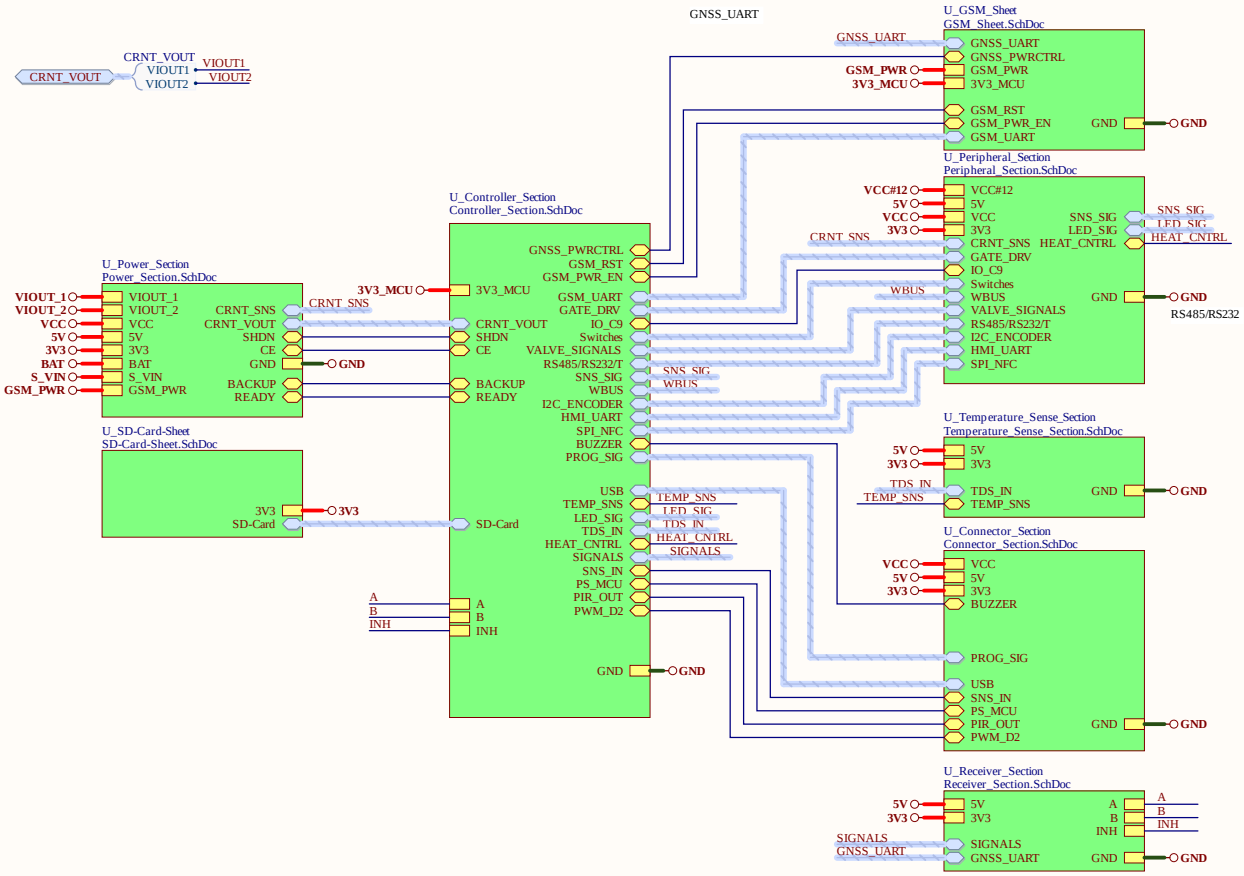
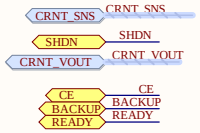
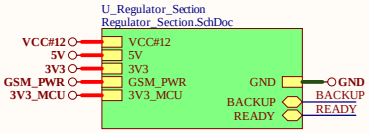
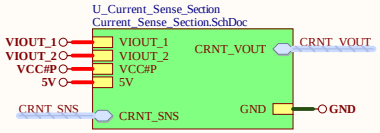


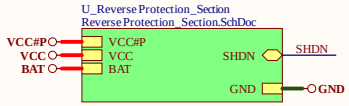
eSpark Consulting Group

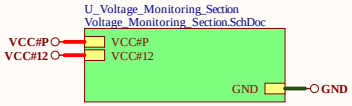
Title: Top_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 27/08/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

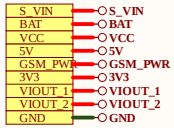


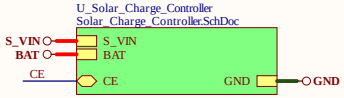








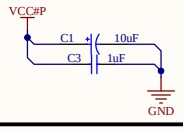




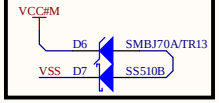
eSpark Consulting Group

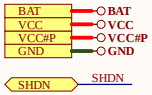
Title: Power_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 14/03/2024	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 17

DECOUPLING CAPACITOR

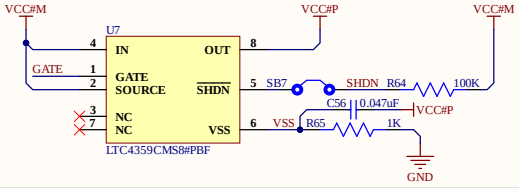


INPUT DIODES

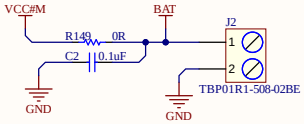




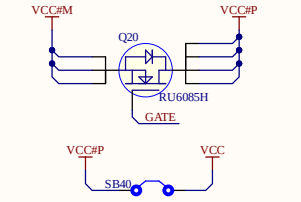
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT

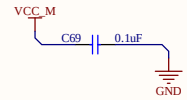


MOSFET

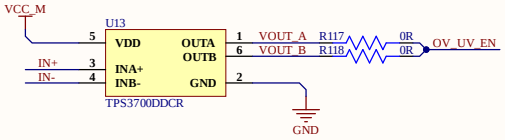


Title: Reverse Protection_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

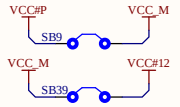
DECOUPLING CAPACITOR



UNDER AND OVER VOLTAGE PROTECTION IC



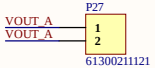
VOLTAGE TO MONITOR



UNDER VOLTAGE OUTPUT

Do Not Mount

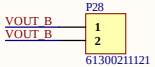
Under Voltage Open Drain Output.
100K Pullup is recommended.

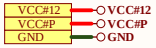


OVER VOLTAGE OUTPUT

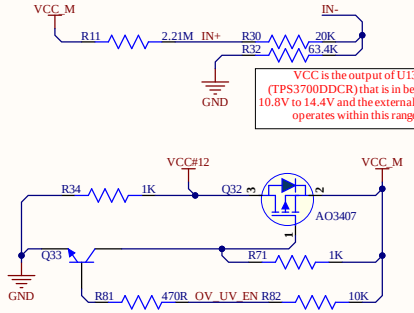
Do Not Mount

Over Voltage Open Drain Output.
10K Pullup is recommended.





SUPPORTING CIRCUITRY



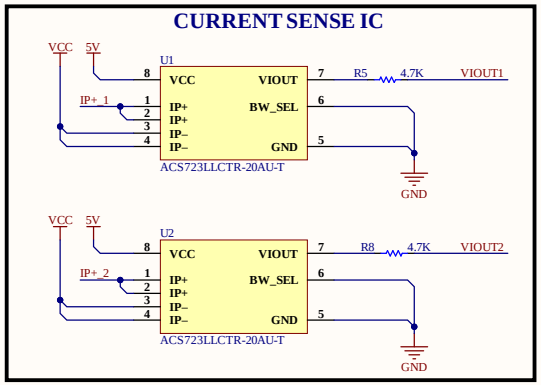
eSpark Consulting Group

Title: Voltage_Monitoring_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2

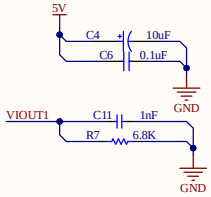
CRNT_VOUT
CRNT_VOUT
VIOUT1
VIOUT2
VIOUT1
VIOUT2

CURRENT SENSE IC

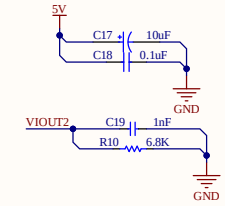


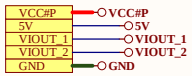
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

VOUT1 SUPPORTING CIRCUIT



VOUT2 SUPPORTING CIRCUIT

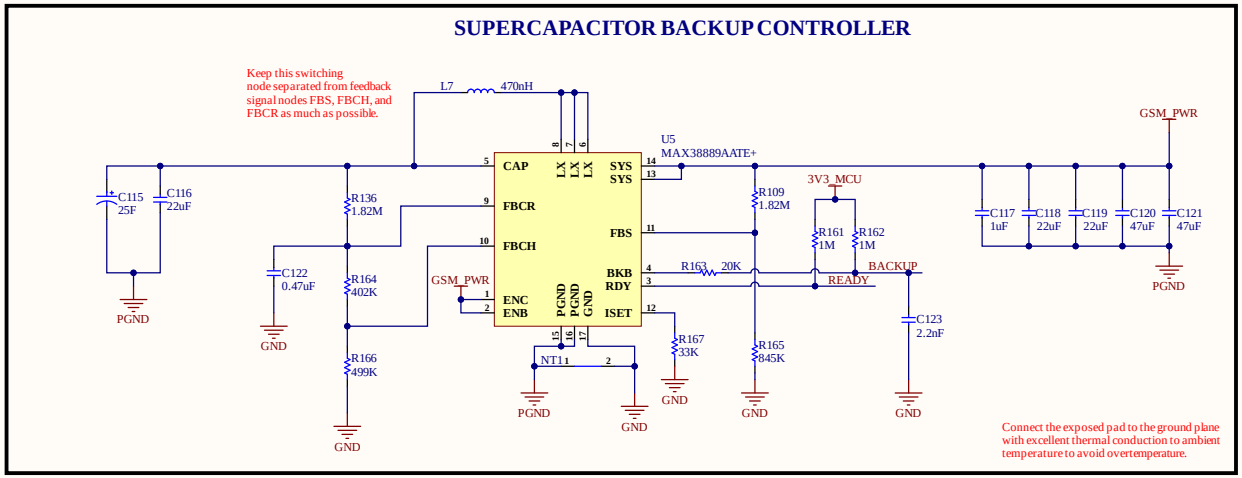




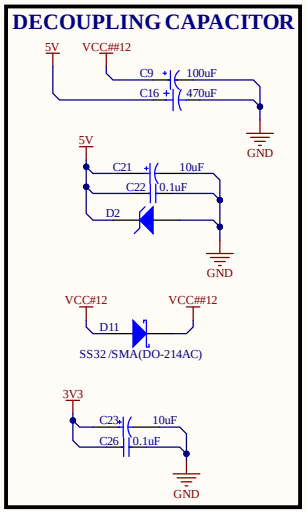
eSpark Consulting Group

Title: Current_Sense_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 14/03/2024	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 17

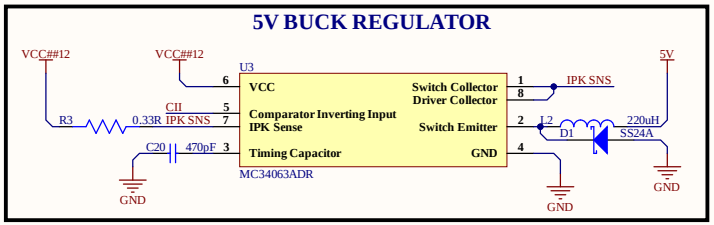
SUPERCAPACITOR BACKUP CONTROLLER



DECOUPLING CAPACITOR

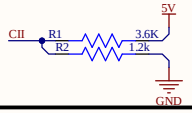


5V BUCK REGULATOR

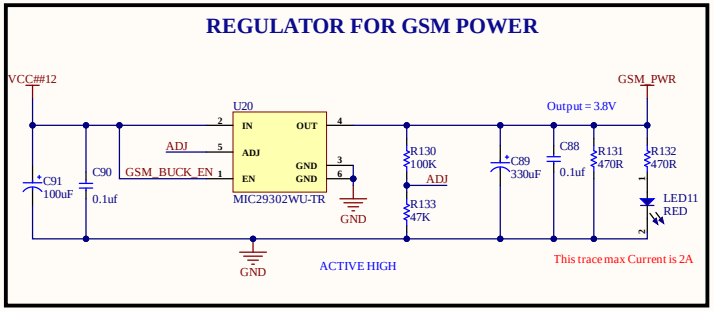


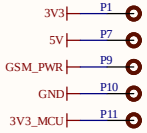
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

SUPPORTING CIRCUIT

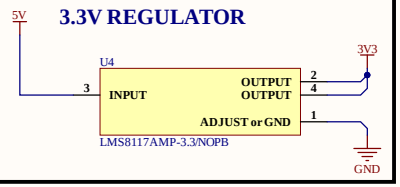


REGULATOR FOR GSM POWER

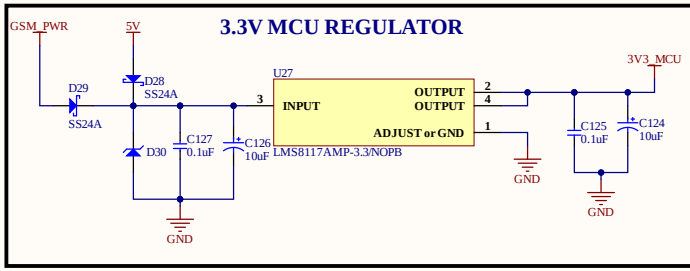


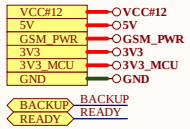


3.3V REGULATOR

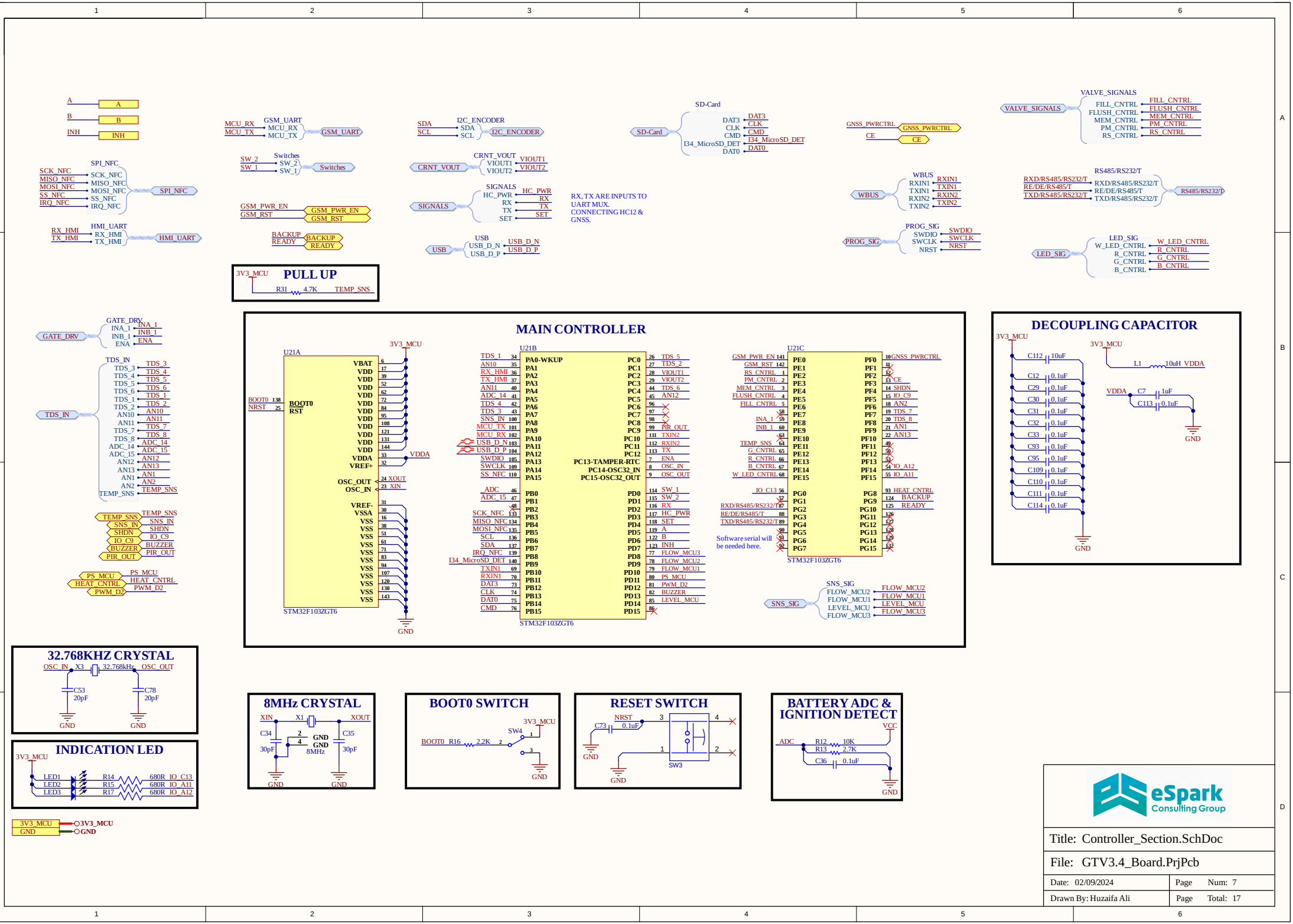


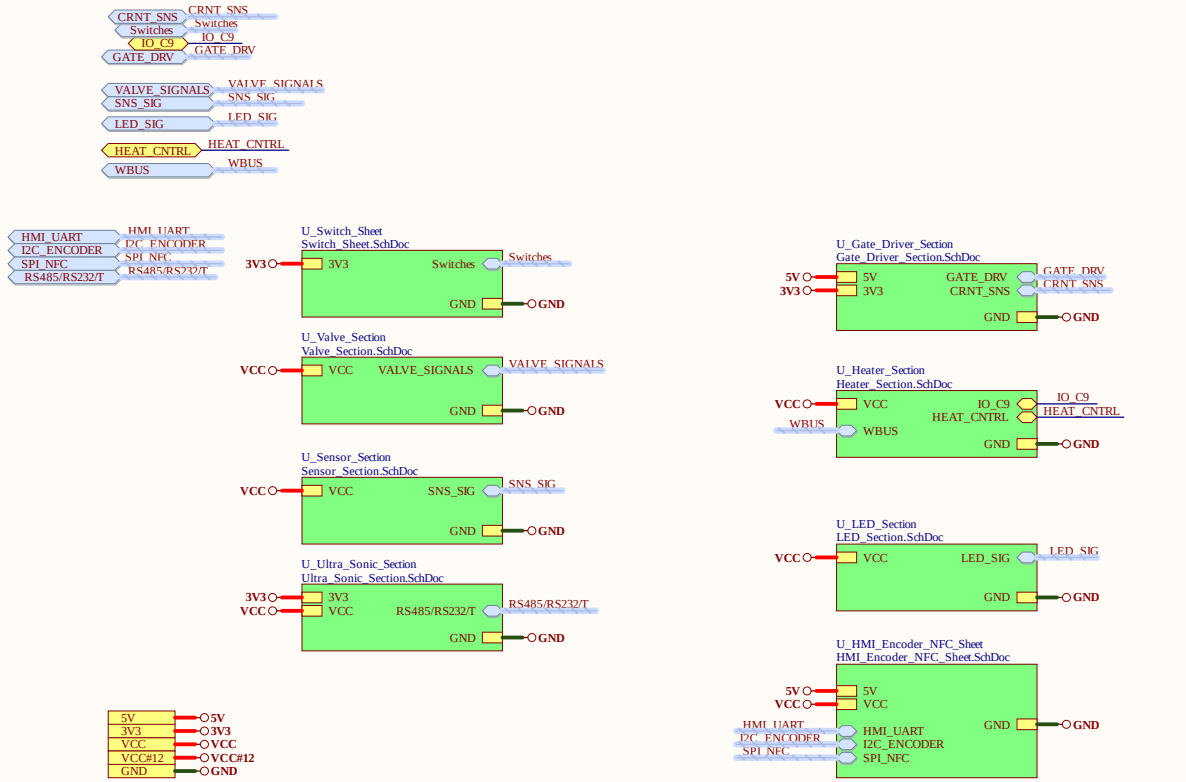
3.3V MCU REGULATOR





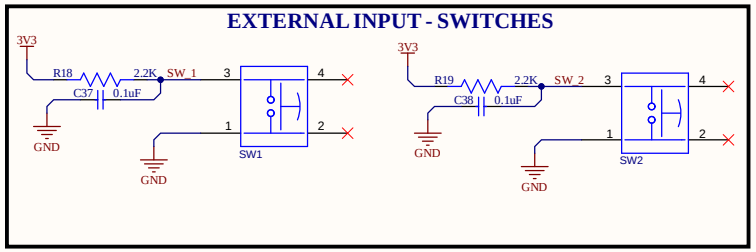
Title: Regulator_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17


A
B
INH
SPI_NFC
SCK_NFC
MISO_NFC
MOSI_NFC
SS_NFC
IRQ_NFC
HMI_UART
RX_HMI
TX_HMI
MCU_RX
MCU_TX
GSM_UART
SW_2
SW_1
Switches
GSM_PWR_EN
GSM_RST
BACKUP
READY
SDA
SCL
I2C_ENCODER
CRNT_VOUT
VIOUT1
VIOUT2
SIGNALS
HC_PWR
RX
TX
SET
USB
USB_D_N
USB_D_P
RX, TX ARE INPUTS TO UART MUX. CONNECTING HC12 & GNSS.
SD-Card
DAT3
CLK
CMD
I34_MicroSD_DET
DAT0
GNSS_PWRCTRL
CE
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
PROG_SIG
SWDIO
SWCLK
NRST
RS485/RS232/T
RXD/RS485/RS232/T
RE/DE/RS485/T
TXD/RS485/RS232/T
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
PULL UP
3V3_MCU
R31
4.7K
TEMP_SNS
GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
PWM_D2
MAIN CONTROLLER
U21A
U21B
U21C
STM32F103ZGT6
BOOT0
NRST
VBAT
VDD
VSS
VDDA
VREF+
VREF-
VSSA
OSC_OUT
OSC_IN
XOUT
XIN
GND
Software serial will be needed here.
FLOW_MCU1
FLOW_MCU2
FLOW_MCU3
LEVEL_MCU
DECOUPLING CAPACITOR
C112
10uF
L1
10uH_VDDA
C12
C29
C30
C31
C32
C33
C93
C95
C109
C110
C111
C114
0.1uF
C7
1uF
C113
32.768KHZ CRYSTAL
X3
32.768kHz
C53
20pF
C78
8MHz CRYSTAL
X1
C34
C35
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R_IO_C13
680R_IO_A11
680R_IO_A12
BOOT0 SWITCH
SW4
BOOT0_R16
2.2K
RESET SWITCH
C73
SW3
BATTERY ADC & IGNITION DETECT
ADC
VCC
R12
10K
R13
2.7K
C36
eSpark Consulting Group
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File: GTV3.4_Board.PrjPcb
Date: 02/09/2024
Page Num: 7
Drawn By: Huzaifa Ali
Page Total: 17



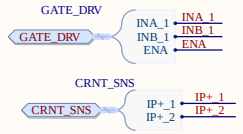
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File:	GTV3.4_Board.PrjPcb
Date: 14/03/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 17

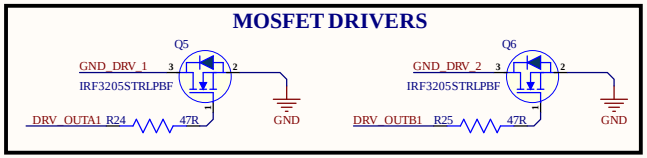


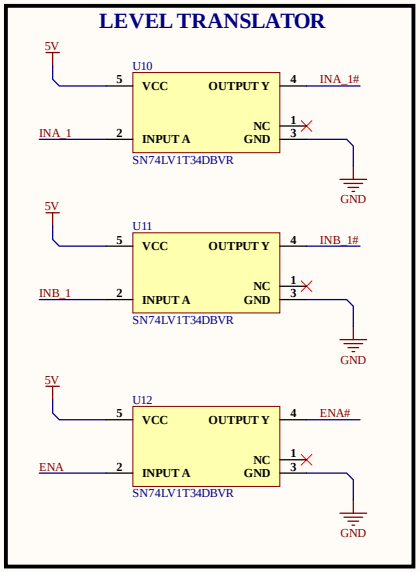


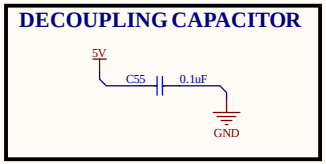


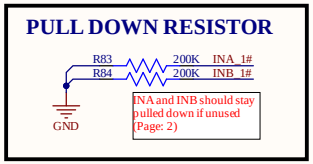
Title: Switch_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 17

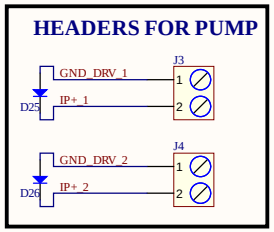


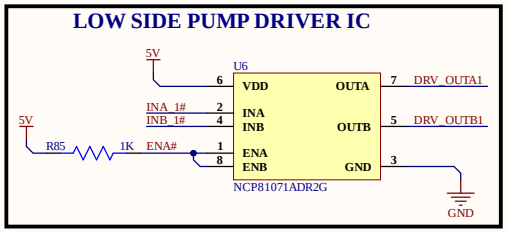


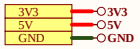




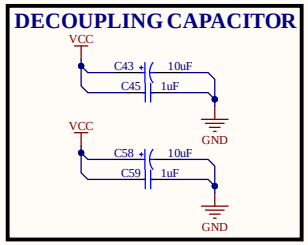






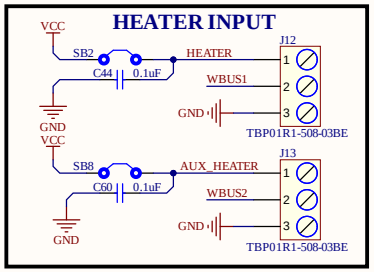


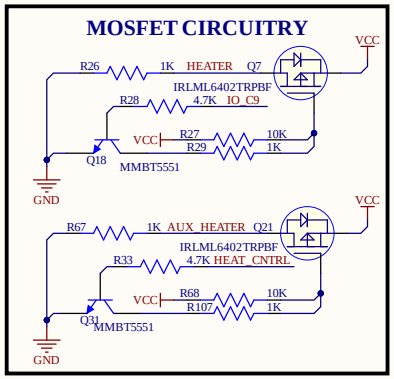
Title: Gate_Driver_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17

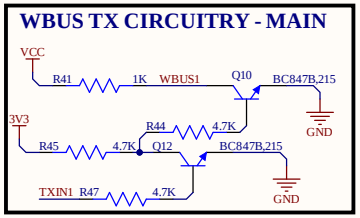


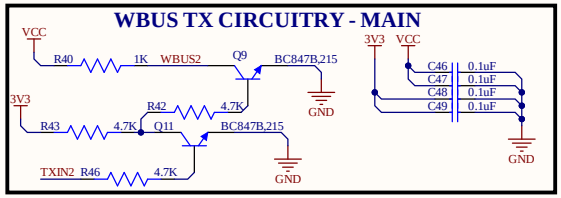


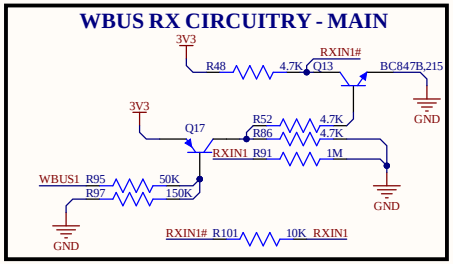


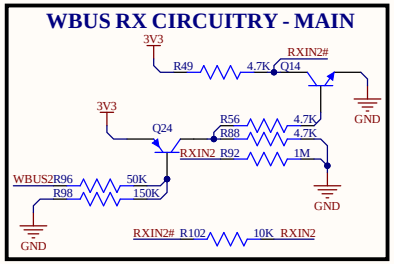


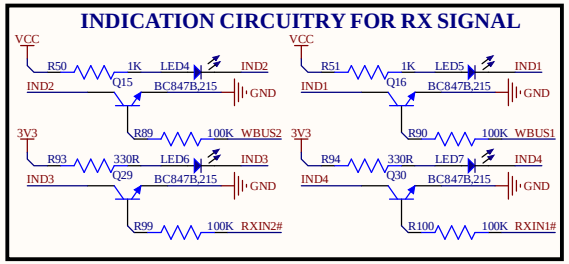








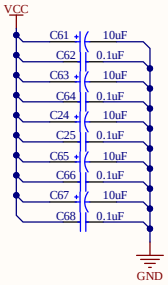




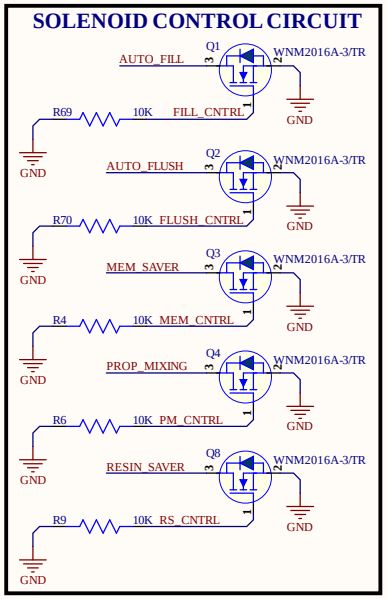
eSpark Consulting Group

Title: Heater_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 14/03/2024	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 17

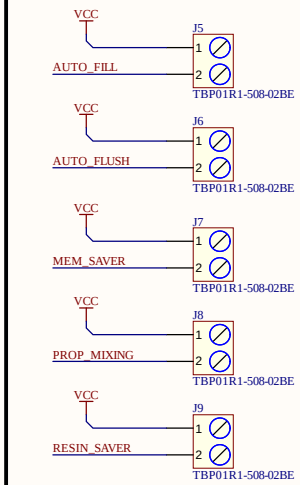
DECOUPLING CAPACITOR



SOLENOID CONTROL CIRCUIT



SOLENOID VALVE INPUTS



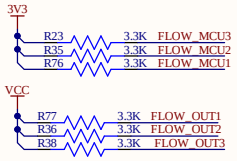




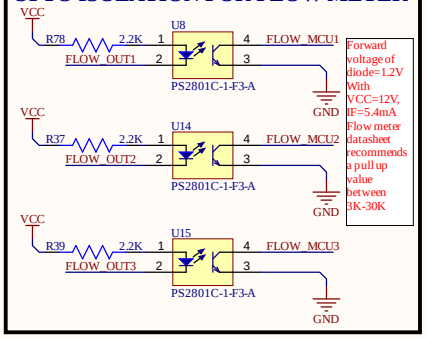
eSpark Consulting Group

Title: Valve_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 14/03/2024	Page	Num: 12
Drawn By: Huzaifa Ali	Page	Total: 17

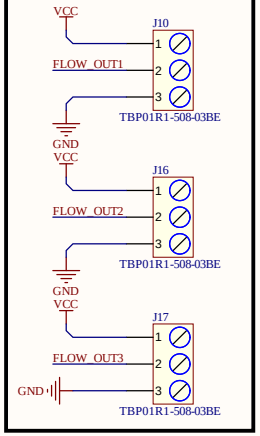
PULL UP RESISTORS



OPTO ISOLATION FOR FLOW METER



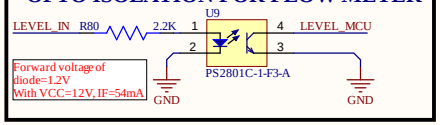
FLOW METER INPUT



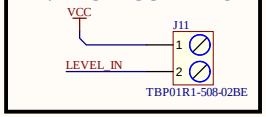
SUPPORTING CIRCUIT

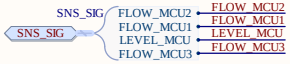


OPTO ISOLATION FOR FLOW METER



LEVEL SENSOR INPUT

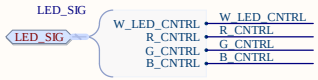


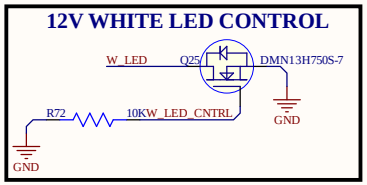


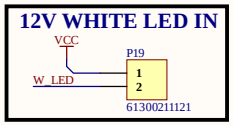


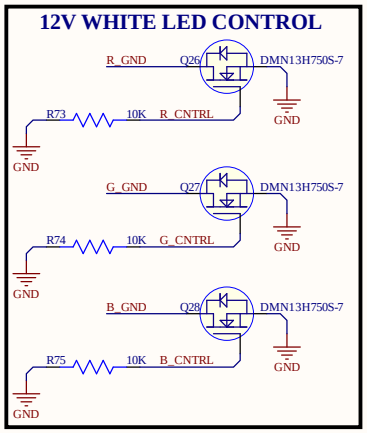
eSpark Consulting Group

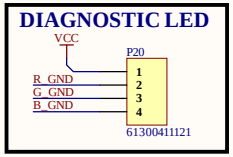
Title: Sensor_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 14/03/2024	Page	Num: 13
Drawn By: Huzaifa Ali	Page	Total: 17







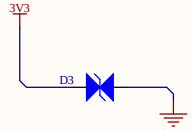




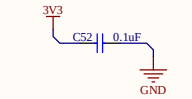


Title: LED_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 17

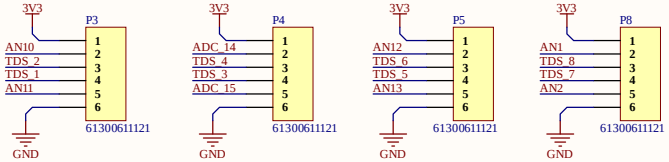
PULL UP RESISTORS



DECOUPLING CAP

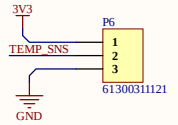


4 X TDS INPUTS

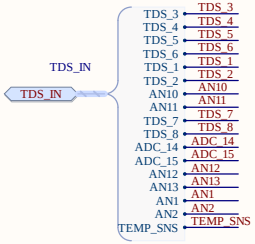


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20



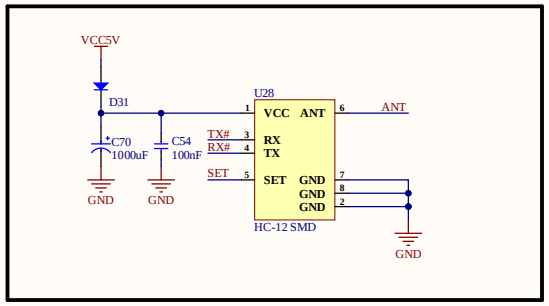




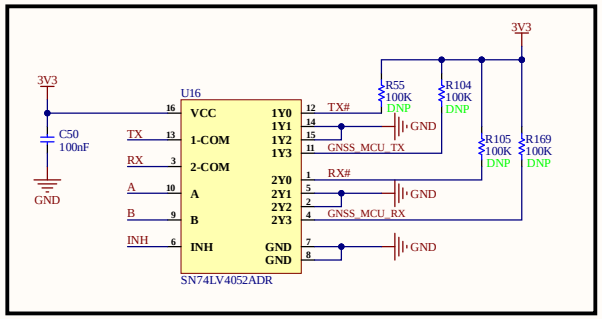
eSpark Consulting Group

Title: Temperature_Sense_Section.SchDoc		
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Date: 14/03/2024	Page	Num: 15
Drawn By: Huzaifa Ali	Page	Total: 17

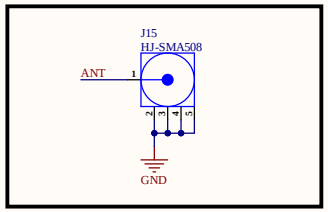
HC-12 433MHz MODULE



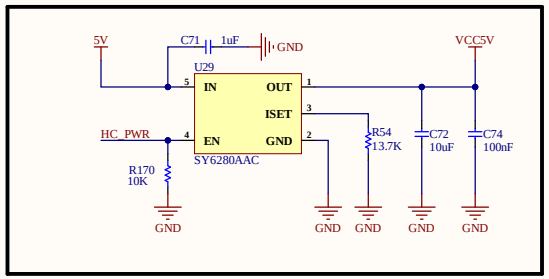
UART MUX

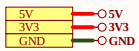


SMA CONNECTOR

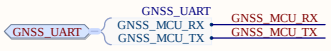


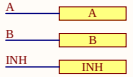
POWER SWITCH





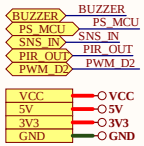


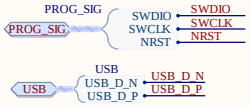


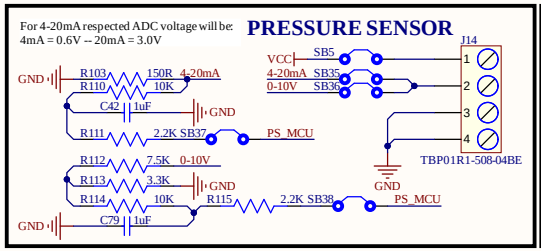


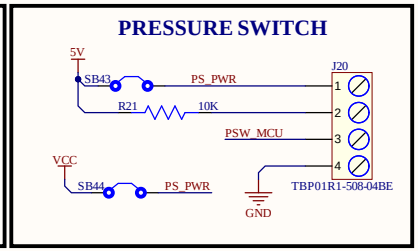
eSpark Consulting Group

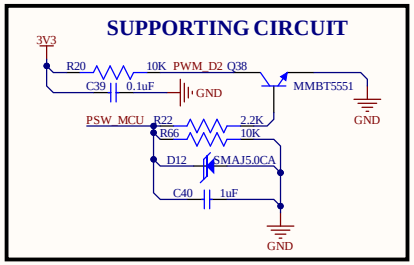
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File: GTV3.4_Board.PrjPcb		
Date: 02/09/2024	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 17

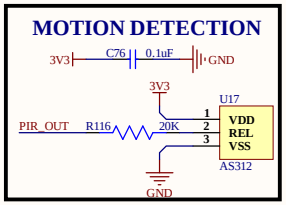


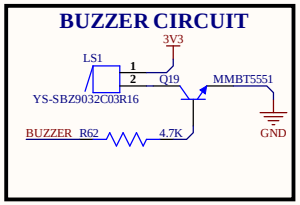


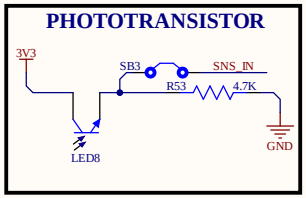


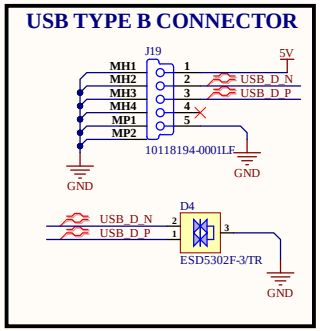


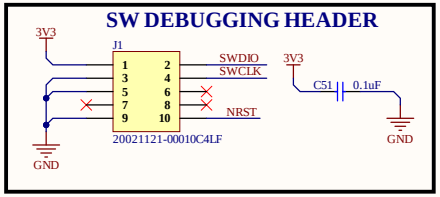






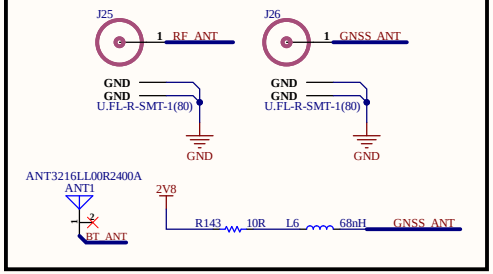




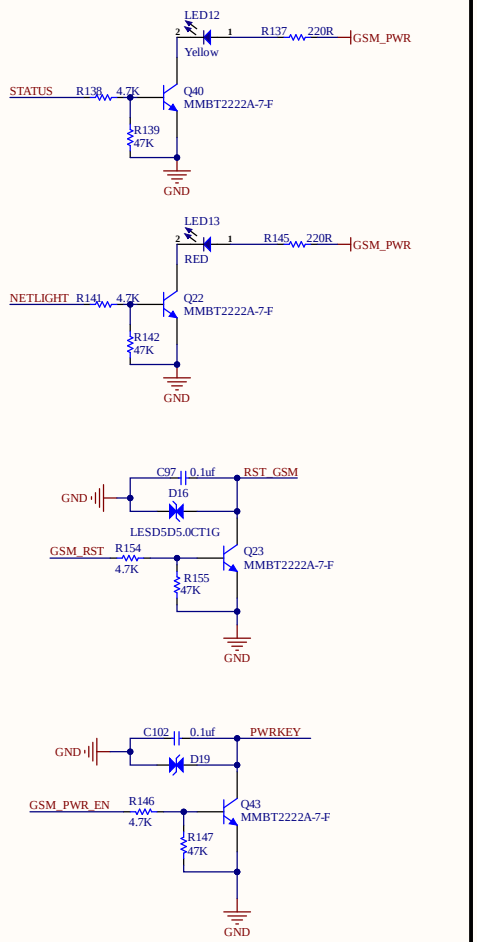


Title: Connector_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: 17
Drawn By: Huzaifa Ali	Page Total: 17

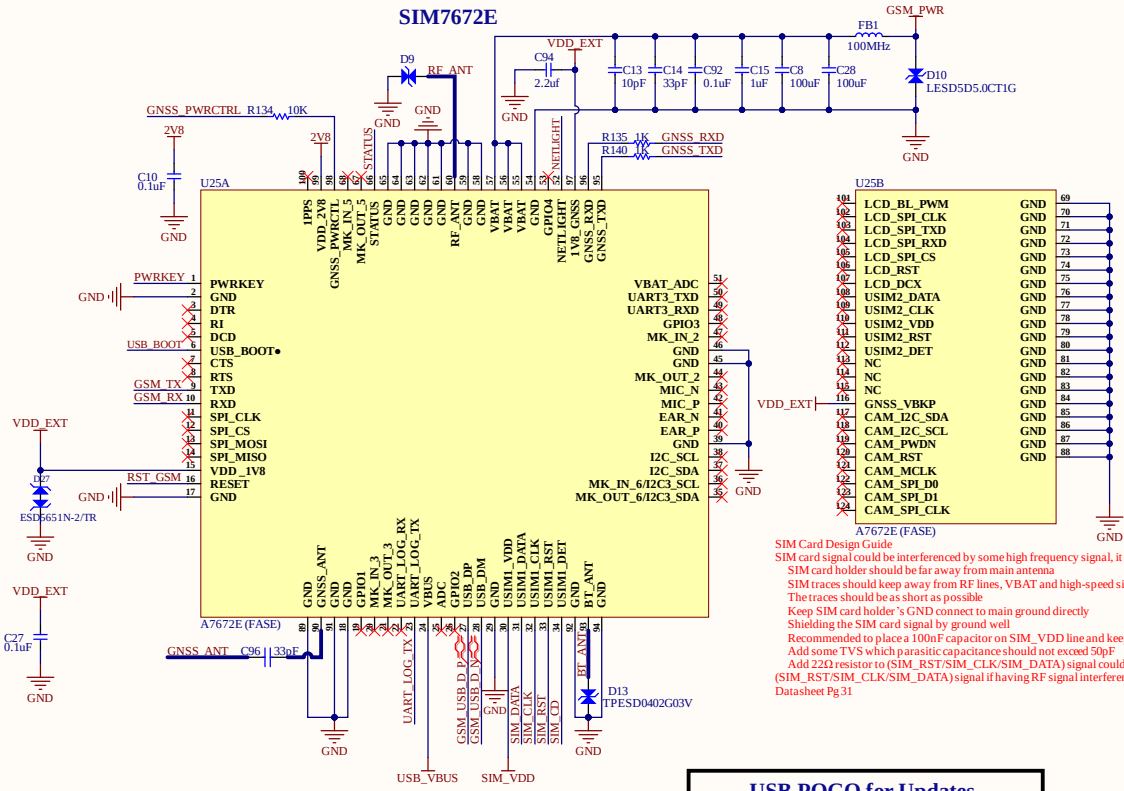
ANTENNA



GSM_COMMANDS



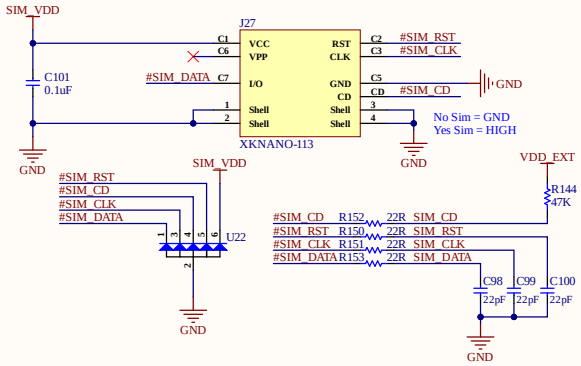
SIM7672E



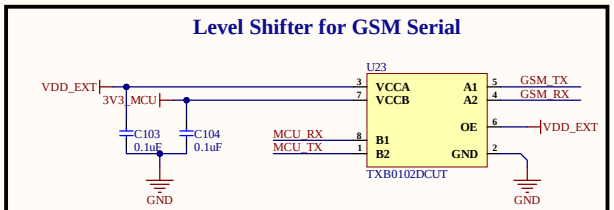
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder 's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

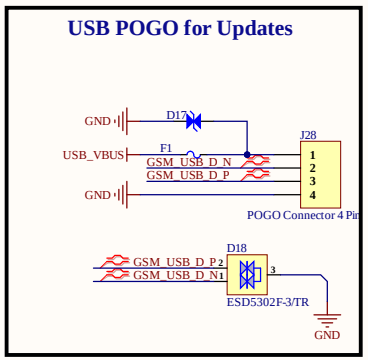
NANOSIM CONNECTOR



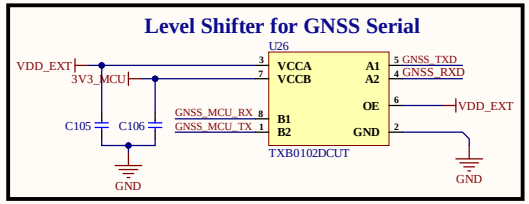
Level Shifter for GSM Serial

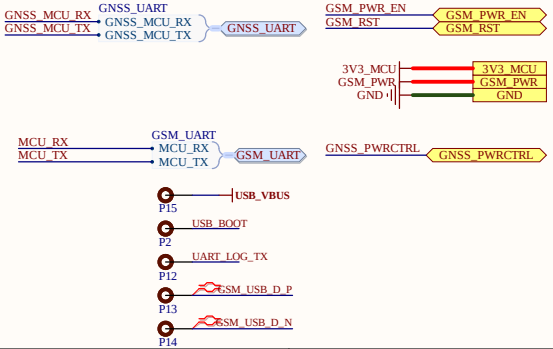


USB POGO for Updates



Level Shifter for GNSS Serial

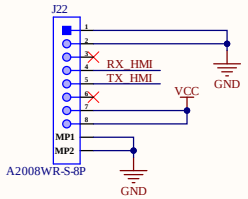




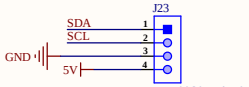
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Date: 14/03/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *

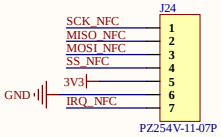
HMI INPUT STONE



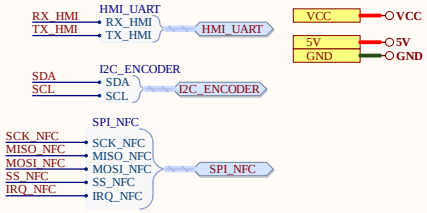
ENCODER INPUT



NFC INPUT

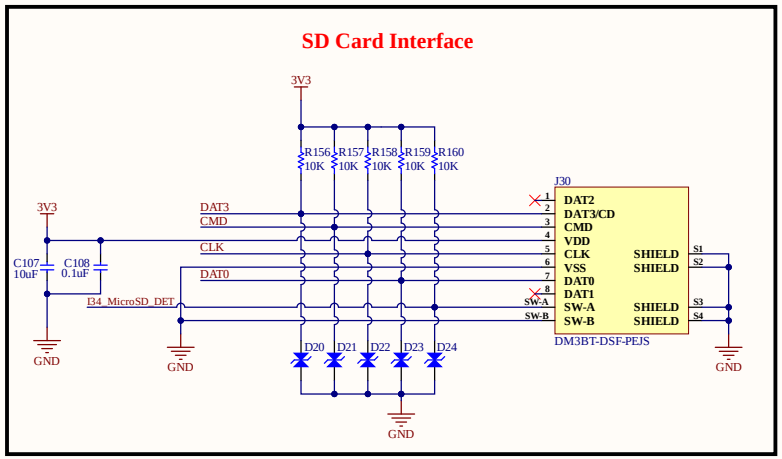


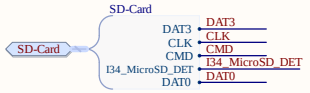
PN532 NFC Module 13.56Mhz



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Title: HMI_Encoder_NFC_Sheet.SchDoc	
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Drawn By: Huzaifa Ali	Page Total: *



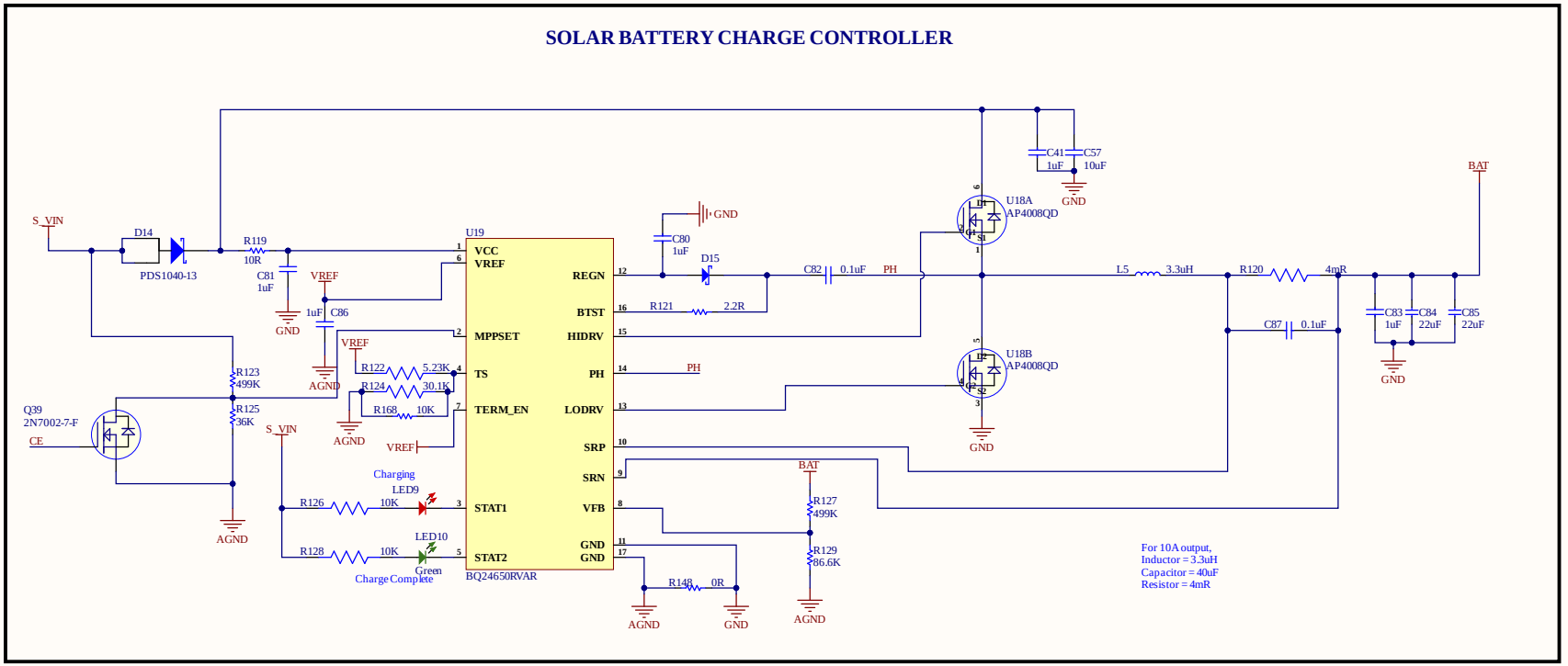




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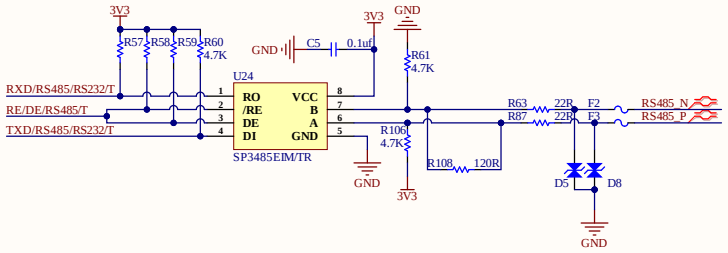
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SOLAR BATTERY CHARGE CONTROLLER

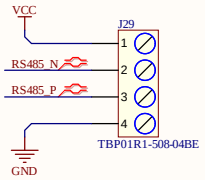


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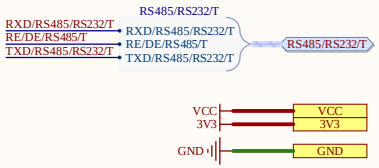
RS485 to UART



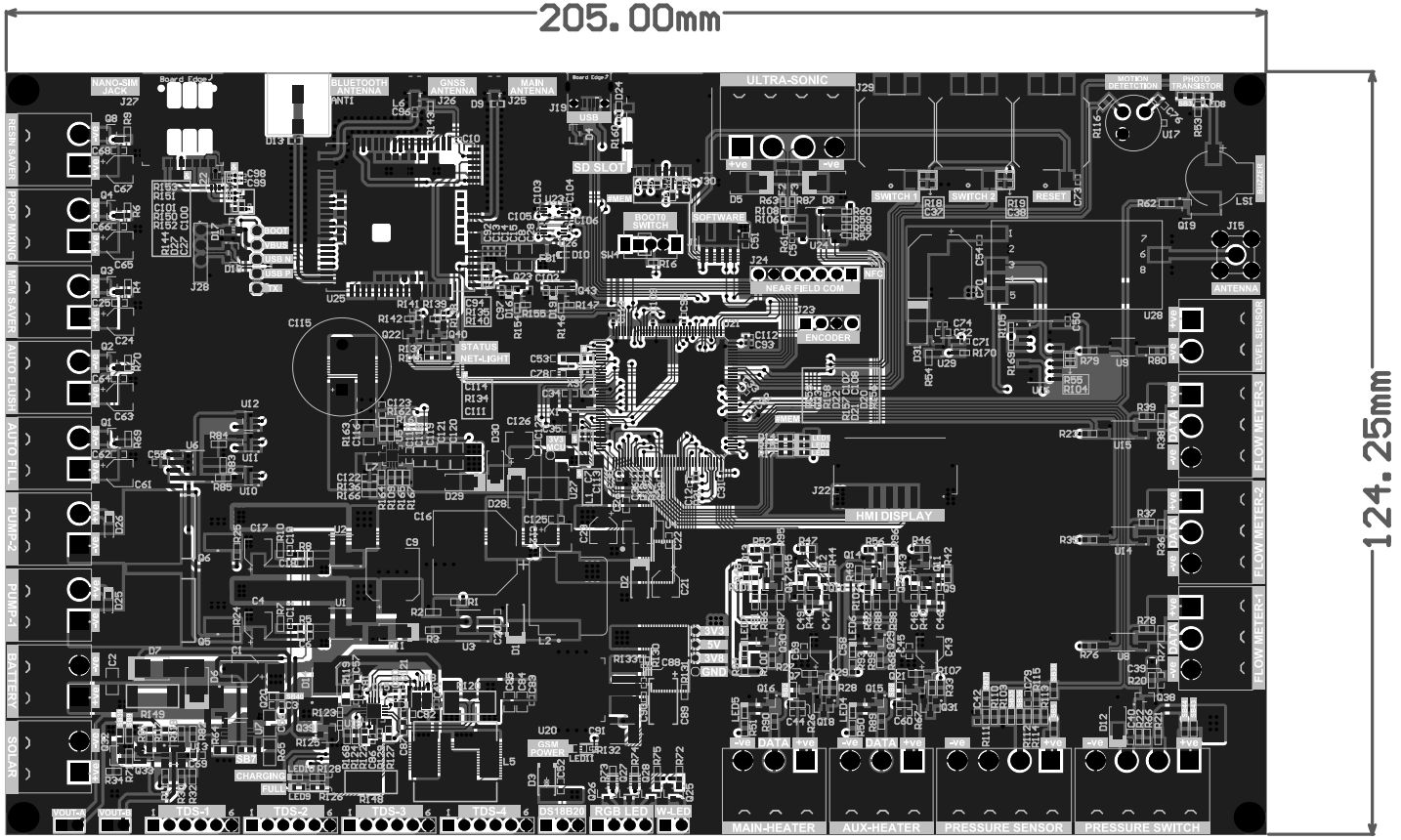
ULTRA-SONIC SENSOR



Part Selected:
https://www.massa.com/industrial/ultrasonic-sensors/#1501873783525-5657d00f-07c9



Title: Ultra_Sonic_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 14/03/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.71mil	3.5	
1	Top Layer		1.38mil		
	Dielectric 2	PP-006	7.99mil	4.4	
2	GND	CF-004	1.18mil		
	Dielectric 1	FR-4	40.55mil	4.6	
3	PWR	CF-004	1.18mil		
	Dielectric 3	PP-006	7.99mil	4.4	
4	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.71mil	3.5	
	Bottom Overlay				
Total board thickness:			63.07mil		